

ColliderAgent revision: audit of open placeholders measurements of 2026-09-16 and proposed text

2026-09-16

Legend

DONE = measured today, text below can be pasted. **PARTIAL** = partly measured; the rest is in progress. **RUNS** = needs the remaining benchmark runs (Opus 4.6 $\times 3$ per benchmark; cross-model). **AUTHOR** = a decision or confirmation only the authors can make.

1 Table S2 (sm.pdf p.4) — software environment

Status: **DONE** for every package (all read inside the images); only the Claude Code version of the *original* runs is open (see note). All versions were read inside the images on zhustation on 2026-09-16 with a probe blueprint (**collider-env-probe**; job logs are cited per row). Two corrections to the current table: the analysis Python stack lives in the *agent runtime*, not in the collider image (the image has no **uproot**), and Delphes is 3.5.1.

Image / runtime	Package	Version
collider	MadGraph5_aMC@NLO	3.7.0
collider	Pythia8	8.316
collider	Delphes	3.5.1
collider	MadAnalysis5	1.11.0 (2025-04-23)
collider	LHAPDF	6.5.5
collider	HepMC	2.06.09
collider	ROOT	6.28/10
collider	Python	3.10.12; numpy 1.26.4, scipy 1.15.3, awkward 2.9.0, matplotlib 3.10.8
mma-het	Wolfram Engine	13.3.0
mma-het	FeynRules	2.3.49 (29 Sep 2021)
micromegas	micrOmegas / CalcHEP	6.3.0 / 3.9.2
agent runtime	Python analysis stack	Python 3.13.12; uproot 5.7.4, awkward 2.9.0, numpy 2.4.4, scipy 1.15.3
agent runtime	Claude Code	2.1.273 (re-runs of 2026-09-16); original runs: see note
agent runtime	LLM	Claude Opus 4.6 (claude-opus-4-6), still served on 2026-09-16

Proposed replacement rows (booktabs, same layout as the current table):

```
collider    & MadGraph5_aMC@NLO & 3.7.0 \\
            & Pythia8      & 8.316 \\
            & Delphes      & 3.5.1 \\
            & MadAnalysis5  & 1.11.0 \\
            & LHAPDF      & 6.5.5 \\
```

```

& HepMC                & 2.06.09 \\
& ROOT                  & 6.28/10 \\
& pyhf                   & 0.7.6 \\
mma-het    & Wolfram Engine   & 13.3.0 \\
            & FeynRules             & 2.3.49 \\
agent runtime & Claude Code    & 2.1.x (AUTHORS: version of the original runs from the usage logs
                    the 2026-09-16 re-runs used 2.1.273) \\
            & LLM                  & Claude Opus 4.6 \\
            & uproot, awkward, numpy, scipy & 5.7.4, 2.9.0, 2.4.4, 1.15.3 \\

```

Note on the Claude Code version: no session log from the April–May runs exists on this machine (the earliest local logs are from August 2026), so the original version has to come from the co-author who ran them (**AUTHOR**). The rows above are otherwise complete.

2 sm.pdf p.5, “LLM inference accounts for a minor fraction [AUTHORS: fill in numbers ...]”

Status: **PARTIAL** — measured for the ALP EFT benchmark with three current models; the Opus 4.6 numbers for all eight benchmarks are being produced now. The claim as written is *not* supported for the light benchmarks: attributing every gap before a model message to inference and every gap before a tool result to tool execution (main session plus stage subagents), the model share of wall-clock was 38% (Opus 5), 54% (Opus 4.8) and 51% (Sonnet 5); time inside `magnus run/status` calls was 60%, 45% and 47% (the two overlap when a stage polls a running job). Proposed wording, to be finalised with the Opus 4.6 numbers:

For the light benchmarks of Table S3 the wall-clock time splits roughly evenly between LLM inference and cluster jobs (model share XX–YY%); for the Dark-SMEFT campaigns, where each campaign submits more than 100 jobs, cluster time dominates.

3 Table S3 (sm.pdf p. 6) — resource usage; also paper.pdf p. 5 and p. 7, reply.pdf p. 4 and p. 6

Status: **RUNS** for the Opus 4.6 column (nine prompts launched 2026-09-16 22:00 UTC at effort `xhigh`; results are appended by `scripts/bench/aggregate.py`). Already measured with the v2 skills, cold memory, one run each:

ALP EFT, $\mathcal{E}_T$ [Fig.2(b)]	Wall [h]	Sub-agents	Magnus jobs	Files	Tokens in [M]	Tokens out [k]	USD
Claude Opus 5	0.87	3	5	17	5.72	37.5	9.06
Claude Opus 4.8	1.05	3	9	16	10.96	43.3	12.55
Claude Sonnet 5	1.38	3	14	24	18.19	34.3	7.54

Definitions to state in the caption: sub-agent calls = **Agent** tool invocations; Magnus jobs = `magnus run` submissions; files written = distinct paths written or edited by the agent (progress records and memory files included); tokens = main session plus all sub-agents, input = uncached + cache-write + cache-read (the CLI’s own usage field covers the main session only and under-reports by about 10×).

4 Table S4 (sm.pdf p. 6; paper.pdf p. 5 “fill in cross-model table or state it as future work”; reply.pdf p. 4 “choose one of the two sentences”)

Status: **PARTIAL**. Recommendation: keep the table with three columns and fill it, because the Referee asked for measurement rather than assertion and the runs are cheap (8–13 USD per attempt). Column choices:

- Opus 4.6 (paper’s model) — three attempts per benchmark, in progress.
- Model B = GPT-5.3-Codex via the Codex CLI, released 2026-02-05, the same day as Claude Opus 4.6 (OpenAI system card dated February 5, 2026; Anthropic Opus 4.6 announcement February 5, 2026). It is the contemporaneous Codex-native model and exercises the skills-only mode (branch `codex-com`). GPT-5.1 (November 2025) predates Opus 4.6 and is not a Codex-tuned model; GPT-5.3-Codex is the better match. To be run by the authors with Codex.
- Model C = Claude Sonnet 5 (or Opus 5) — one attempt already succeeded on the ALP benchmark; a same-vendor smaller model answers the “does a cheaper model follow the skills” question the Referee raised.

Reply sentence to keep: “We have therefore run the four main-text benchmarks with Claude Opus 4.6 (three attempts each), Claude Sonnet 5 and GPT-5.3-Codex (via Codex) and report the success rates in Table S4 of the SM.” Delete the “not been able to complete” alternative once the runs are in.

5 Table S5 (sm.pdf p. 6, text p. 6 “fill in the per-benchmark counts and failure modes”)

Status: **RUNS**. Three sandboxed attempts per benchmark with Opus 4.6: the first attempt of each is running; two more are one command each (`scripts/bench/run_benchmark.sh <arxiv> <fig> claude-opus-4-6 --effort xhigh`). Failure modes are recorded in `verdict.yaml` per run with the paper’s four categories; a human confirms each verdict. S5 is worth keeping: it is the only place that quantifies the “occasional failed reproductions” sentence on p. 6, and the sandbox pipeline the SM describes (clean workspace, one prompt) is exactly what the harness does.

6 Other markers

Where	Marker	Status	Proposed action
paper.pdf p. 5	“skills-only mode executes the hello-world workflow of Sec. S-III [confirm that this has been tested]” (Codex/Cursor)	AUTHOR	The Codex-compatible layout exists (branch <code>codex-com: .agents/skills, .codex/agents, AGENTS.md</code>). Run the hello-world prompt with Codex on that branch once and keep the sentence; otherwise weaken to “is designed to”.
paper.pdf p. 7, Table I caption	“tokens consumed and wall-clock time [fill in]”	RUNS	Same numbers as Table S3 (Opus 4.6 column); cite Table S3 instead of repeating them.
paper.pdf p. 9–10	“[confirm the extent of human intervention between campaigns]”	AUTHOR	Dark-SMEFT campaigns were not re-run; statement must come from the campaign logs.

Where	Marker	Status	Proposed action
paper.pdf p. 10; reply.pdf p. 6	“several weeks of work for an experienced phenomenologist [adjust estimate]”	AUTHOR	Judgement call; unchanged by measurements.
reply.pdf p. 1	checklist (1)–(4): fill S3/S4/S5, confirm human intervention, set <code>\markupfalse</code> , remove red markers	PARTIAL	(1) in progress, (2) author, (3)–(4) mechanical once (1)–(2) are done.
reply.pdf p. 4	“the LLM token consumption [fill in from the usage logs]”	RUNS	From Table S3; note the main+sub-agent definition.
reply.pdf p. 4	“The wall-clock time of a benchmark (1–6 hours) ..”	DONE	Consistent with today’s runs (0.9–1.4 h for the lightest benchmark); keep.
sm.pdf p. 4	“archived release ... DOI and per-run provenance packaged as Workflow Run RO-Crates [confirm or remove]”	AUTHOR	No RO-Crate export exists in the repository today; either remove or keep as a plan.
sm.pdf p. 5	Table S4 placeholder text “e.g. Claude Opus 4.6 vs. Claude Sonnet, GPT-5.x via Codex, Gemini via ADK”	PARTIAL	Replace with the three columns chosen above; drop Gemini/ADK unless the python-agent path is run.
sm.pdf p. 6	Table S3 Dark-SMEFT rows (sub-agent calls, tokens)	AUTHOR	Not reproducible here; from the campaign logs.

7 Provenance of today’s numbers

Branch `feat/skill-evolution-v2` of the ColliderAgent repository: `docs/paper-tbd-audit-2026-09-16.md` (measurements and observations), `bench_runs/<label>/metrics.json` and `verdict.yaml` per run, `scripts/bench/` for the tables, probe blueprint `collider-env-probe` on zhustation for the image versions. Skills used for the re-runs are the v2 skills (context-lean rewrite of 2026-09-16), not the April versions; the Opus 4.6 re-runs therefore measure the current system.